

Lab 2 (Part 1): Forensic Bias Audit

Deliverable 1: Forensic Evidence Pack

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Step 2: Individual Reliability and Counterfactual Audit

This audit was conducted using Google PAIR's What-If Tool (Google PAIR, n.d.), loaded with its built-in COMPAS-like recidivism demo classifier, to probe individual reliability near the 0.5 decision threshold and group-level fairness across race.

Monotonicity Trial 1 — Datapoint 342 (African-American, Female, age 27)

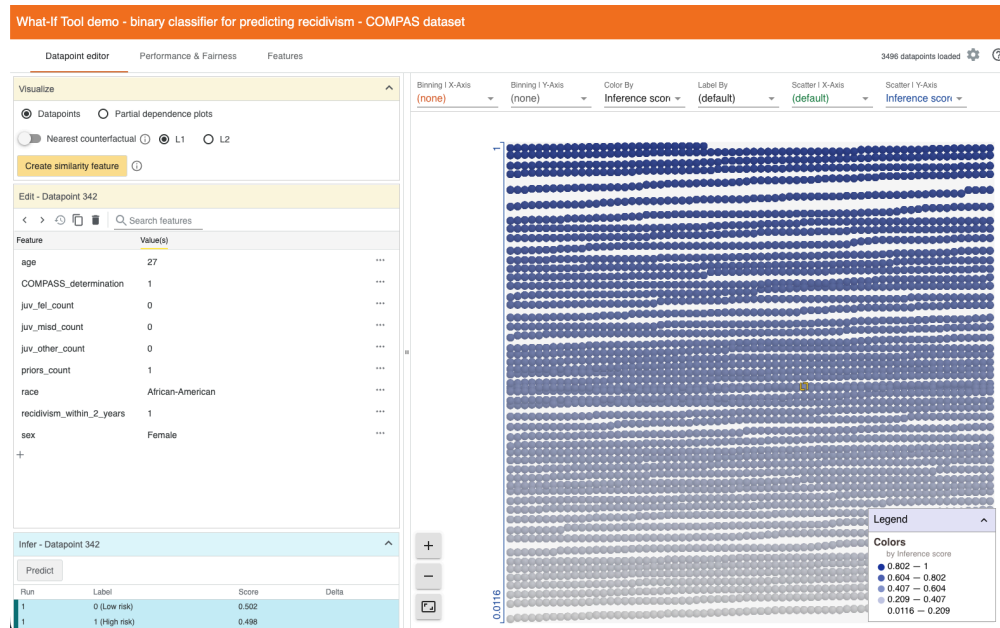
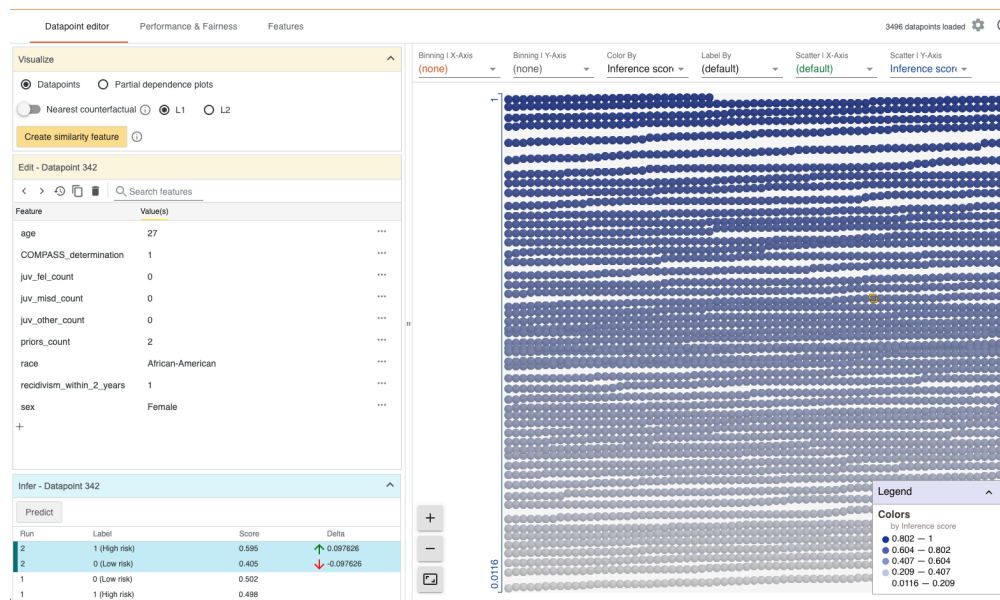


Figure 1a. Datapoint 342 before edit: priors_count = 1, score 0.502/0.498 → Low Risk.



Deliverable 1: Forensic Evidence Pack

Figure 1b. Datapoint 342 after `priors_count` edited 1 $\rightarrow$ 2: score 0.405/0.595 $\rightarrow$ High Risk ($\Delta +0.098$). Label flip observed.

Monotonicity Trial 2 — Datapoint 464 (Caucasian, Male, age 22)

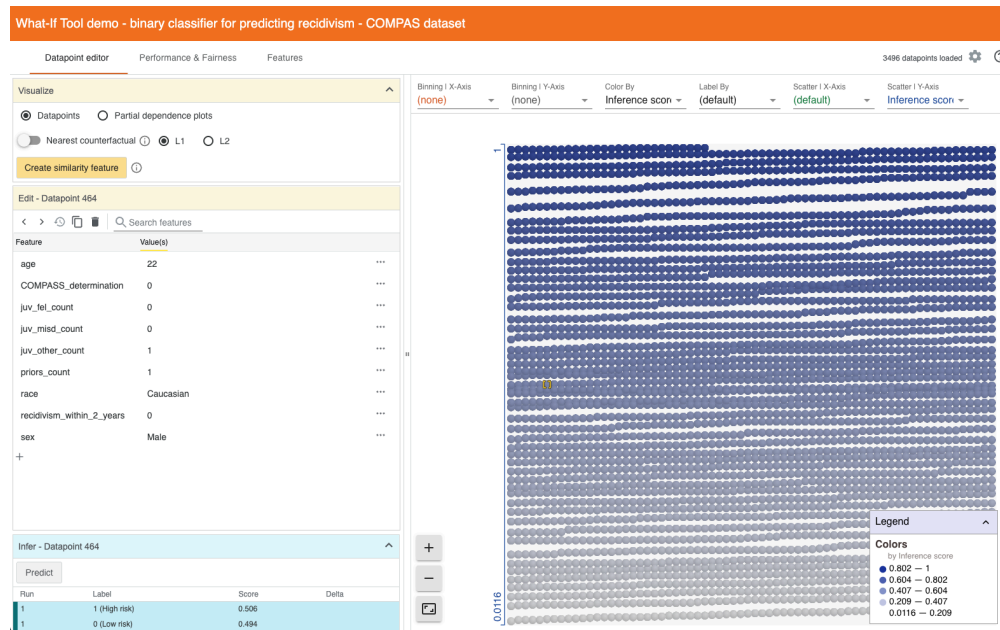


Figure 2a. Datapoint 464 before edit: `priors_count` = 1, score 0.494/0.506 $\rightarrow$ High Risk.

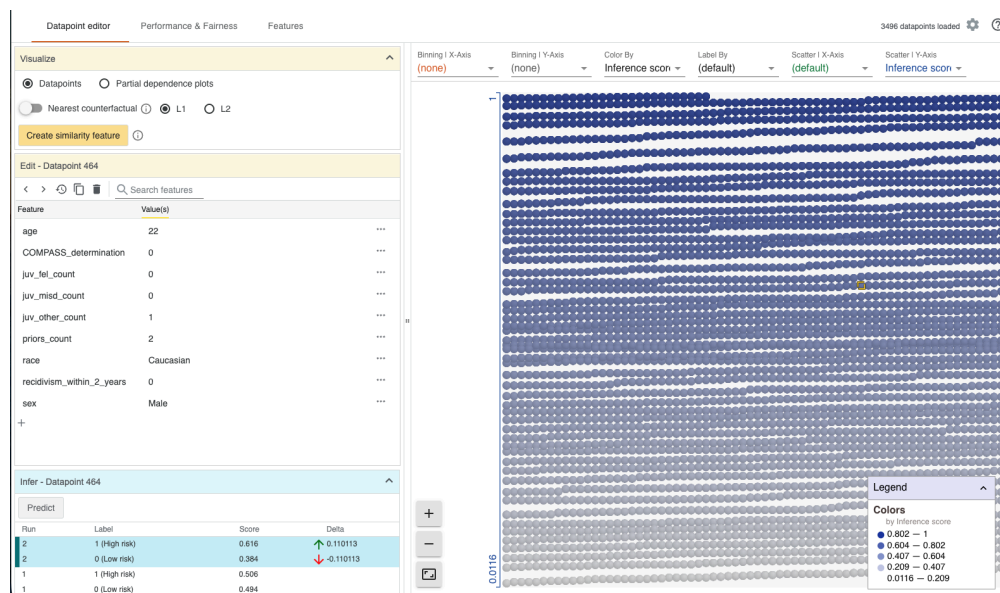


Figure 2b. Datapoint 464 after `priors_count` edited 1 $\rightarrow$ 2: score 0.384/0.616 $\rightarrow$ High Risk ($\Delta +0.110$). No label flip; score moved further from threshold.

Deliverable 1: Forensic Evidence Pack

Monotonicity Trial 3 — Datapoint 894 (Caucasian, Female, age 23)

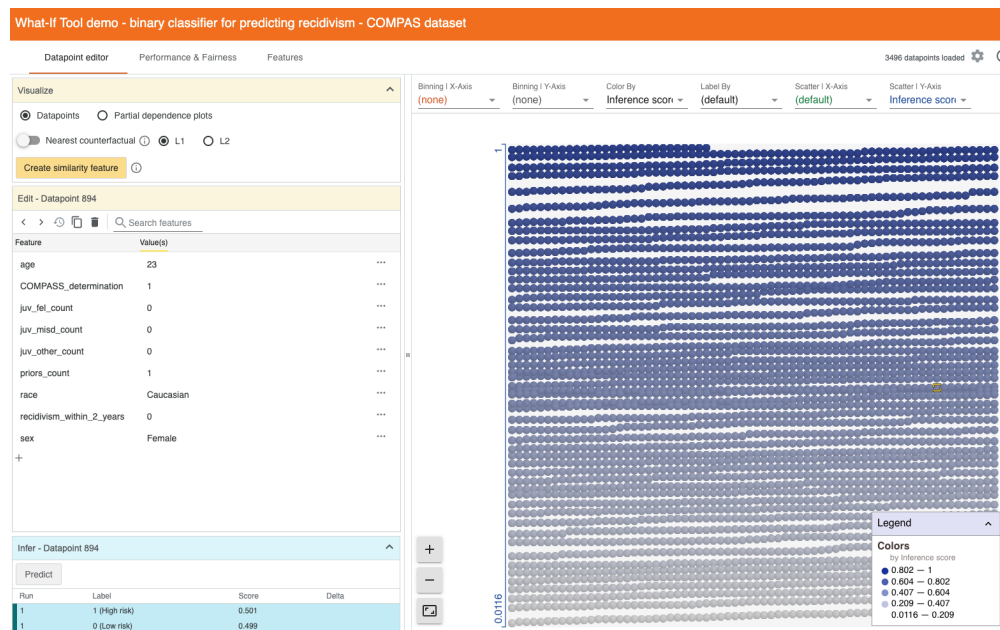


Figure 3a. Datapoint 894 before edit: $priors_count = 1$, score $0.499/0.501 \rightarrow$ High Risk.

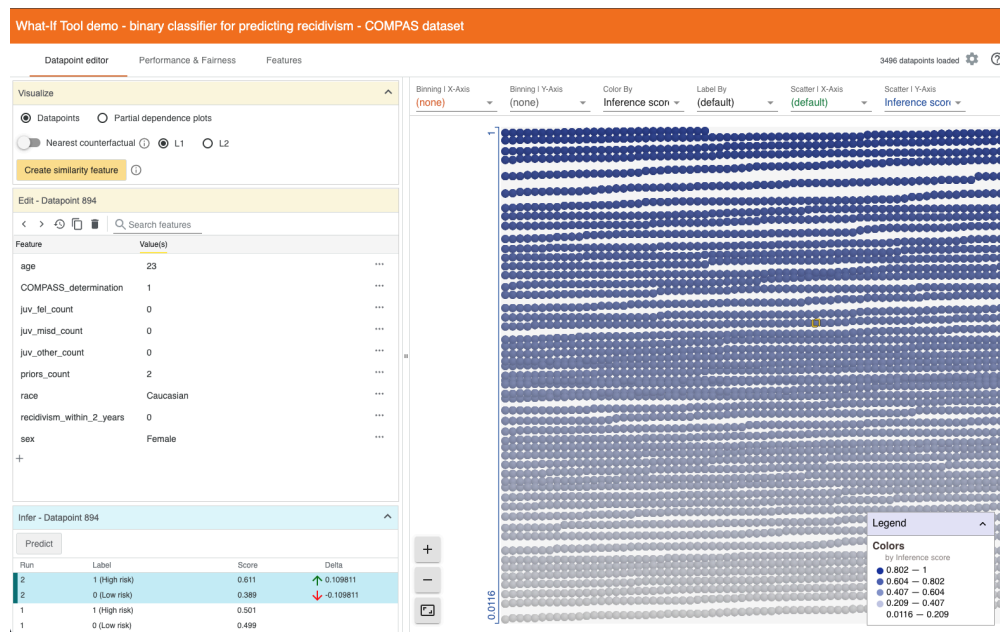


Figure 3b. Datapoint 894 after $priors_count$ edited $1 \rightarrow 2$: score $0.389/0.611 \rightarrow$ High Risk ($\Delta +0.109$). No label flip; score moved further from threshold.

Counterfactual Sensitivity — Datapoint 464 vs. Nearest Counterfactual (Datapoint 2047)

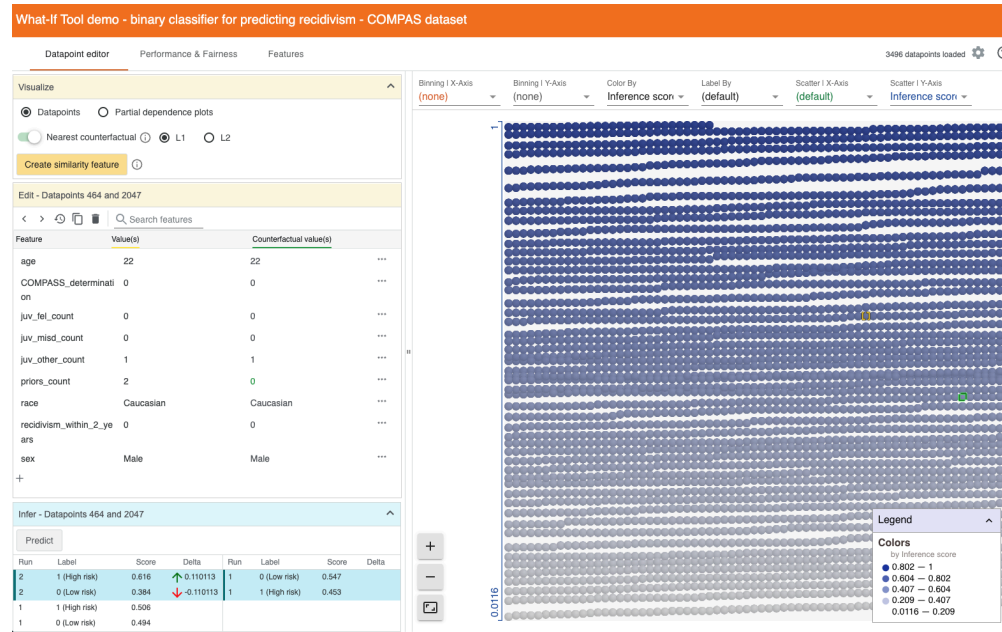


Figure 4. Nearest counterfactual (L1 distance) for Datapoint 464 (*priors_count* = 2, High Risk, 0.616) is Datapoint 2047 (*priors_count* = 0, Low Risk, 0.453/0.547). *priors_count* is the only differing feature; age, race, sex, and juvenile counts are identical.

Summary of Step 2 Probing Trials

Datapoint	Group / Sex / Age	Prior# (start)	Initial Score (Label)	Prior# (new)	New Score (Label)	Δ Score	Flip?	Monotonic?
342	African-American / F / 27	1	0.498 (Low Risk)	2	0.595 (High Risk)	+0.098	Yes	Yes
464	Caucasian / M / 22	1	0.506 (High Risk)	2	0.616 (High Risk)	+0.110	No	Yes
894	Caucasian / F / 23	1	0.501 (High Risk)	2	0.611 (High Risk)	+0.109	No	Yes

Table 1. Summary of the three monotonicity trials in Section 2.2, for quick reference alongside the full screenshots and interpretation above.

Forensic Interpretation

Across the three near threshold trials examined (Datapoints 342, 464, and 894), boundary fragility was substantial. All three original inference scores fell within 0.008 of the 0.5 decision threshold (0.498, 0.506, and 0.501, respectively), meaning the model's High Risk / Low Risk determination for these individuals rested on a probability margin smaller than the resolution of a single categorical feature like *priors_count*. Across the three near-threshold trials examined (Datapoints 342, 464, and 894), boundary fragility was substantial. All three original inference scores fell within 0.008 of the 0.5 decision threshold (0.498, 0.506, and 0.501, respectively), meaning the model's High Risk / Low Risk determination for these individuals rested on a probability margin smaller than the resolution of a single categorical feature like *priors_count*. Datapoint 342 went above the threshold entirely after the model got an extra prior conviction added, so a person went from "Low Risk" (0.498) to "High Risk" (0.595). This is one example that shows the fragility of the system when it comes to defendants near the cut-off, and a binary decision made by a classifier has real-life consequences (such as a person getting detained before

trial or bail being denied) and this decision is being made based on a single incremental fact that adds to one feature rather than on an overall stable and long-profile assessment of risk.

Monotonicity held in all three trials, increasing `priors_count` from 1 to 2 raised the inference score in every case (+0.098, +0.110, and +0.109 respectively), with no instance of a score decrease or reversal. No non-monotonic behavior was observed, so `priors_count` behaves as expected in isolation, meaning more prior convictions increase predicted risk. However, monotonic direction alone does not imply stability. The magnitude of these increases, roughly ten percentage points from a single-unit feature change, is large relative to the threshold's tolerance. A feature behaving correctly in direction can still produce an unstable classification outcome for anyone whose baseline score sits near 0.5, which is precisely the population a pretrial risk tool is most often asked to adjudicate.

Counterfactual sensitivity was tested on Datapoint 464 in its post-edit state (`priors_count` = 2, High Risk, score 0.616), since that was the active configuration when the toggle was applied. Using L1 distance, WIT returned Datapoint 2047 as the nearest oppositely labeled point (Low Risk, score 0.453). Of nine recorded features, `priors_count` was the only value that differed between the two records, dropping from 2 to 0; age (22), race (Caucasian), sex (Male), and all three juvenile count fields were identical. For this defendant, the model's closest opposite outcome neighbor is separated by a single criminal history value rather than by age, juvenile history, or race, indicating the local decision boundary is being driven by `priors_count` rather than a demographic proxy. This finding is instance specific and should not be read as evidence the model is free of disparate impact at the group level; it does, however, reinforce Section 2.2's result that `priors_count` alone carries enough weight near the threshold to move a defendant across the classification line.

Step 3: Group Fairness Audit

Baseline Fairness Table (Custom Thresholds, 0.5 for both groups)

Group	Count	Base Rate	FPR	FNR	Accuracy	F1
African-American	1,904	53.5%	24.5%	11.4%	64.1%	0.70
Caucasian	1,111	42.8%	9.6%	23.7%	66.7%	0.54

Note that the underlying base rates differ substantially (53.5% vs. 42.8%). Per Kleinberg et al.'s (2016) impossibility theorem, this difference in outcome prevalence mathematically guarantees that a shared baseline threshold (0.5) cannot simultaneously achieve equalized error rates (FPR/FNR) and within-group calibration establishing that the baseline disparities documented above are a structural statistical constraint, not a hyperparameter tuning flaw. See Deliverable 2, Section 2 for the full theoretical treatment.

Post-Equal Opportunity Fairness Table

Group	Count	Threshold	FPR	FNR	Accuracy	F1
African-American	1,904	0.59	18.0%	19.0%	63.1%	0.65
Caucasian	1,111	0.39	18.2%	15.1%	66.7%	0.62

Ethical Tax Calculation

Ethical tax = Baseline accuracy – Equal Opportunity accuracy

Group accuracies were weighted by group count (1,904 African-American / 1,111 Caucasian) to obtain a comparable overall figure at each stage, since WIT does not display an unsliced overall accuracy in the race-sliced view.

Baseline weighted accuracy: 65.06% Equal Opportunity weighted accuracy: 64.43%

Ethical tax = 0.63 percentage points

$D_{\text{fair}} (|FPR_{\text{AA}} - FPR_{\text{C}}| + |FNR_{\text{AA}} - FNR_{\text{C}}|)$ fell from 27.2 at baseline to 4.1 under Equal Opportunity — an 85% reduction — for a cost of under one point of accuracy.

Note: True Positive Rate and False Negative Rate are complementary ($TPR = 1 - FNR$), so Equal Opportunity's requirement of equal TPR across groups is mathematically equivalent to requiring equal FNR across groups — the two are used interchangeably below.

Interpretation

The Equal Opportunity constraint cost roughly 0.6 points of overall accuracy (65.1% → 64.4%, weighted by group count) while cutting the combined error-rate disparity from 27.2 to 4.1 points, an 85% reduction. That is a strikingly favorable exchange rate: a small predictive cost purchased a large fairness gain, achieved by raising the African-American threshold from 0.5 to 0.59 and lowering the Caucasian threshold from 0.5 to 0.39.

In other words and with respect to the harmful allocation, this change mainly transfers the burden of classification error through the society without wiping it out completely. For instance, elevating the African-American cutoff reduced the False Positive Rate from 24.5% to 18.0%; so fewer African-American defendants who didn't reoffend were classified as High Risk incorrectly, but the True Positive Rate also dropped from 88.6% to 81.0%. The public safety trade off is real, some actual reoffenders who would have been flagged under baseline thresholds are no longer flagged. Conversely, the Caucasian group's TPR rose from 76.3% to 84.9%, catching more actual reoffenders at the cost of a higher FPR. The governance cost, then, is a substitution of error types across groups rather than a reduction in total error, achieved for a strikingly small accuracy penalty a trade most institutions could defend, but one that still requires explicitly choosing which group's false positives or false negatives to prioritize reducing.

References

Google PAIR. (n.d.). *What-If Tool* [Software]. <https://pair-code.github.io/what-if-tool/>

Kleinberg, J., Mullainathan, S., & Raghavan, M. (2016). Inherent trade-offs in the fair determination of risk scores. *arXiv*. <https://arxiv.org/abs/1609.05807>

AI Assistance: Used Grammarly for minor editing only